CCCA Meta-Atlas — Bassez et al. 2021 (Breast)

Breast Cancer · 10x · 65,885 malignant cells → 65,615 analysed (41 samples ≥20 cells)

Do malignant cells recapitulate embryonic-development programs? Geometry: non-branching (linear) · Gates: GO / GO / NA

GO / NO-GO GATES

Gate 1 mixing & continuity — **GO**

θ=10 Harmony on sample: iLISI 4.9→8.6/41 (p10 3.2). θ-sweep on the embedding: θ=2 fragments (45% main), θ=10 one continuous body (>99.9%). Lobes span all 41 patients (max 0.17). Complete island removal (0.17% specks). → GO.

Gate 2 module geography — **GO (non-branching)**

Gast~pSom co-vary (raw −0.10 / sm −0.21), both oppose Crest (sm −0.24 / −0.82). Along pseudotime Gast/pSom fall (ρ −0.12/−0.27), Crest rises (ρ +0.30); joint pSom&Crest 0.23<1 → inverse terminal gradient. One axis, no arms. No inversion.

Gate 3 cross-repression — **NA**

Non-branching (curve 2 tips/0 forks; PPT 0 Gast-enriched tips). No bifurcation → NA.

PARAMETER CHOICES & RATIONALE

Threshold	≥20 cells/sample → 41/42 kept (gap-finding; anchor ≥100 drops 39 real samples).
Embedding	n_pcs=30 (scree, centres 20–40); Harmony θ=10 (iLISI plateau + continuous embedding).
Islands	single-pass radius-graph specks (0.17%); no re-embedding (avoids runaway removal).
Trajectory	scFates curve, N=20 (2 tips, 0 forks, fold-free self-approach 2.6).
Root	tip 2 = Gast/pSom-high, Crest-low developmental origin (Crest is late, never root).
Geometry	non-branching / linear (Gast/pSom→Crest single axis; corr x_lin~t = +0.94).
EMT-TFs	detection-gated panel incl. FOXC/PRRX → ZEB2 (root) ↔ PRRX1 (terminus).

STEP FIGURES

Verdict: the embryonic pattern reproduces as one continuous (Gast/pSom)→Crest axis — a linear (non-branching) manifold. GO / GO / NA.

